

QANDEEL

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Herd, don't harvest: live-release management of Blue Blubber jellyfish swarms after ENEC's early warning

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We keep swarms out of the intake and return them to the sea alive

Problem

Blue Blubber swarms (30–45 cm, 1–3 kg) drift onto seawater intakes. Clearing them takes heavy manual work and almost none survive.

Solution

After ENEC’s alert, software plans the response; a bubble curtain at the breakwater gap holds the swarm out; a boom with a closed-bottom retention bag gathers it in the water; it is released alive down-current.

Evidence

Working simulation on real 2025 Copernicus currents and ERA5 wind; dashboard; 16 automated tests; open code.

Next

A 1:20 tank test (AED 300 kit) to measure what the model assumes, then a sea pilot with ENEC.

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swarms that truly arrived were covered, in a hindcast of 100 alerts on real 2025 Gulf data

77%

fewer curtain running hours than switching on for every alert

88–100%

of jellyfish kept by the boom with our retention bag (model), vs 0–54% for an open oil-spill boom

46%

of the energy of a fixed airflow, by setting air hour by hour from real currents and wind

A large, local, hard-to-handle jellyfish

ENEC already detects swarms offshore. The open question is what to do next: manage high volumes with less manual work and less harm.

The target: Blue Blubber

- One of the two dominant outbreak species on the UAE coast; reproduces year-round (Aravind 2024)
- 30–45 cm bell, ~1–3 kg each (Yang et al. 2018 mass formula)
- 87–100% sit in the top 1 m by day (Pitt & Kingsford)
- Large adults swim >10 cm/s: not just passive drifters

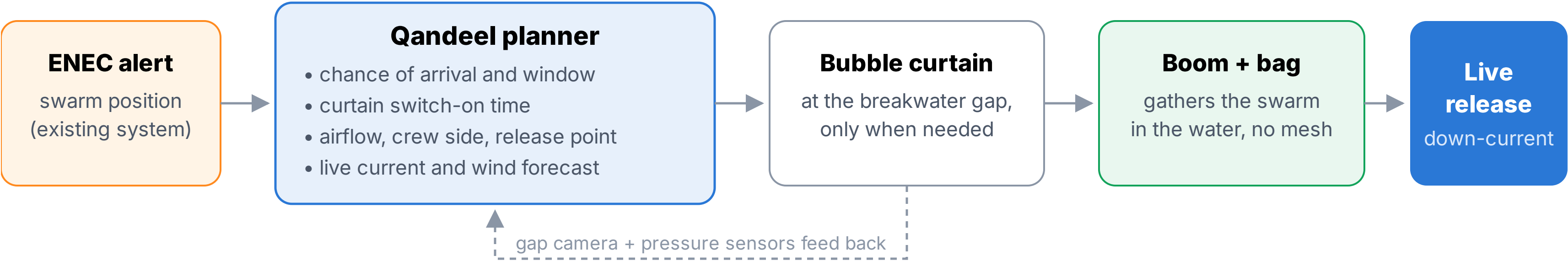
What happens at intakes

- Swarms follow the intake flow through the breakwater opening and pile on screens
- Worldwide this has forced power reductions and shutdowns (e.g. Torness 2011, Oskarshamn 2013)
- In the region, ~300 t of jellyfish clogged intakes in Oman

Why it is hard

- Manual raking and trucking, sometimes for days
- Jellyfish crushed on screens or discarded: almost none survive
- Cutting robots (e.g. JEROS) shred them into fragments
- No management method has ever been tested on this species

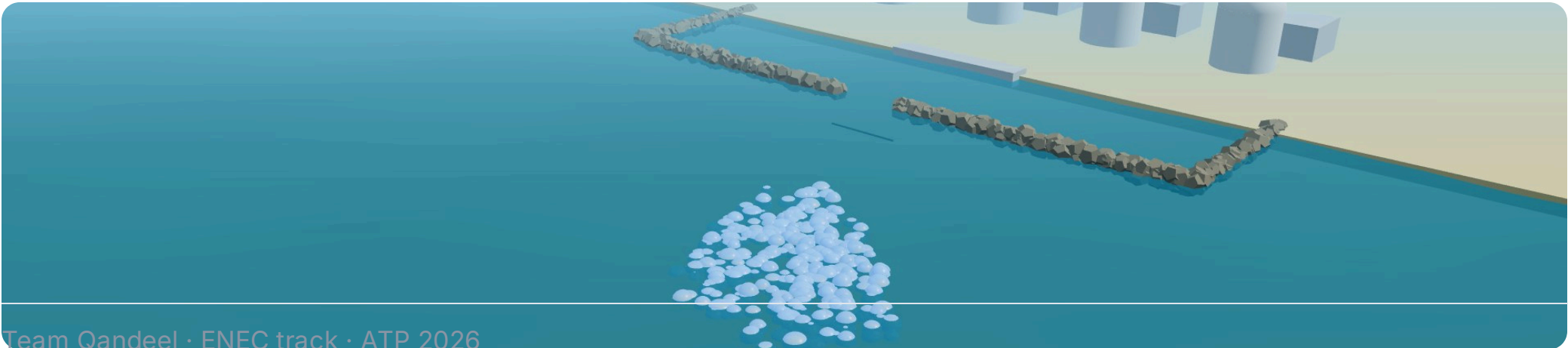
Guard the one gap, move the water, keep every jellyfish in the sea



1 • The gap is a chokepoint
Seawater, and the swarm, can only reach the intake through one opening.

2 • Move the water, not the animal
A rising-bubble current holds them out; nothing is cut, pumped or netted.

3 • Borrow proven oil-spill tools
Booms and boats exist today; we found and fixed why they leak jellyfish.



The breakwater leaves one opening to the intake. A swarm offshore must pass through it, so that is the one place we act. Image: our 3D model (stylised; not to scale).

HOW IT WORKS

One swarm, from alert to live release

- 1

Alert.

ENEC reports a swarm, e.g. 15 km NNW. The planner receives it automatically.
- 2

Plan.

400 simulated drifts of 60 swimming jellyfish each, on the live current and wind forecast: 54% chance of arrival, window 10–77 h.
- 3

Prepare.

Curtain on at +7 h; crew staged on the approach side; release point chosen.
- 4

Deflect.

Curtain across the gap holds the swarm outside; airflow follows the current.
- 5

Gather.

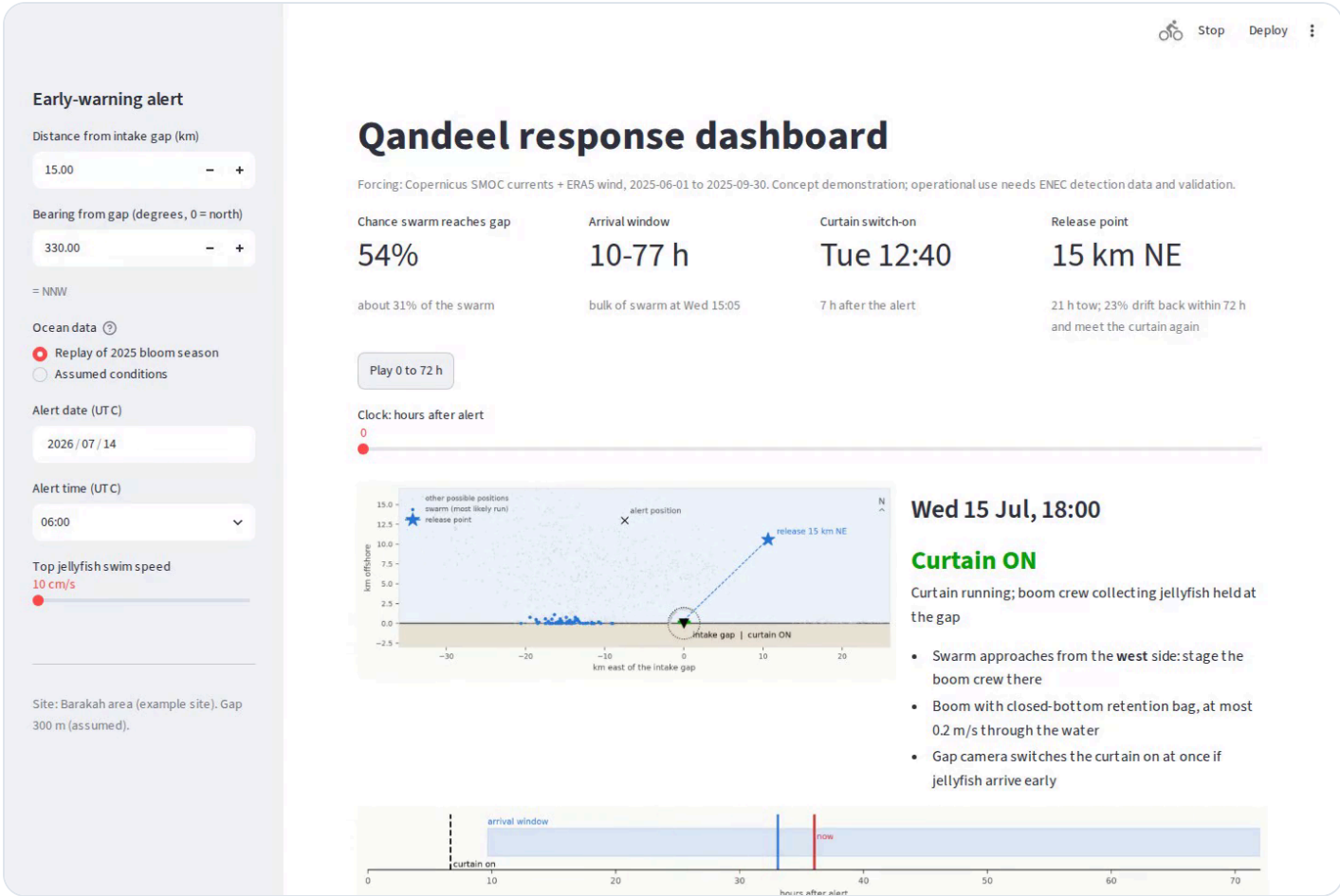
Two boats close a U-boom with a closed-bottom retention bag around them.
- 6

Release.

Move with the current at ≤ 0.2 m/s through the water; open the boom down-current.
- 7

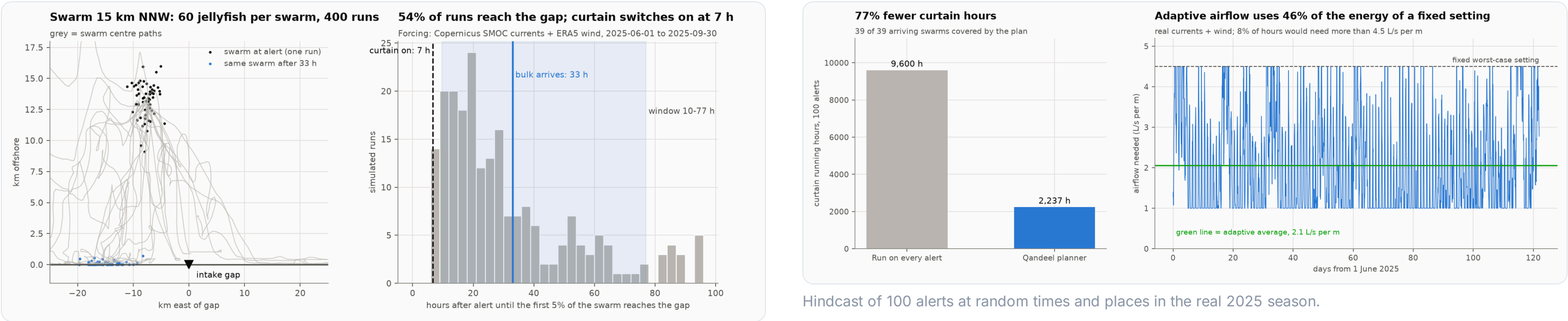
Learn.

Camera and sensor data improve the next plan. Backup: the gap camera starts the curtain at once if jellyfish arrive early.



Qandeel operator dashboard (working prototype, Streamlit): chance of arrival, window, switch-on clock time, swarm map, curtain status, release point. Data: Copernicus SMOC currents + ERA5 wind, summer 2025.

The planner runs the curtain only when it matters



Swarm 15 km NNW: 400 runs on real June–September 2025 conditions.

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Every swarm that truly reached the gap was covered by the plan; the curtain stayed off for 45 of 61 that did not arrive.

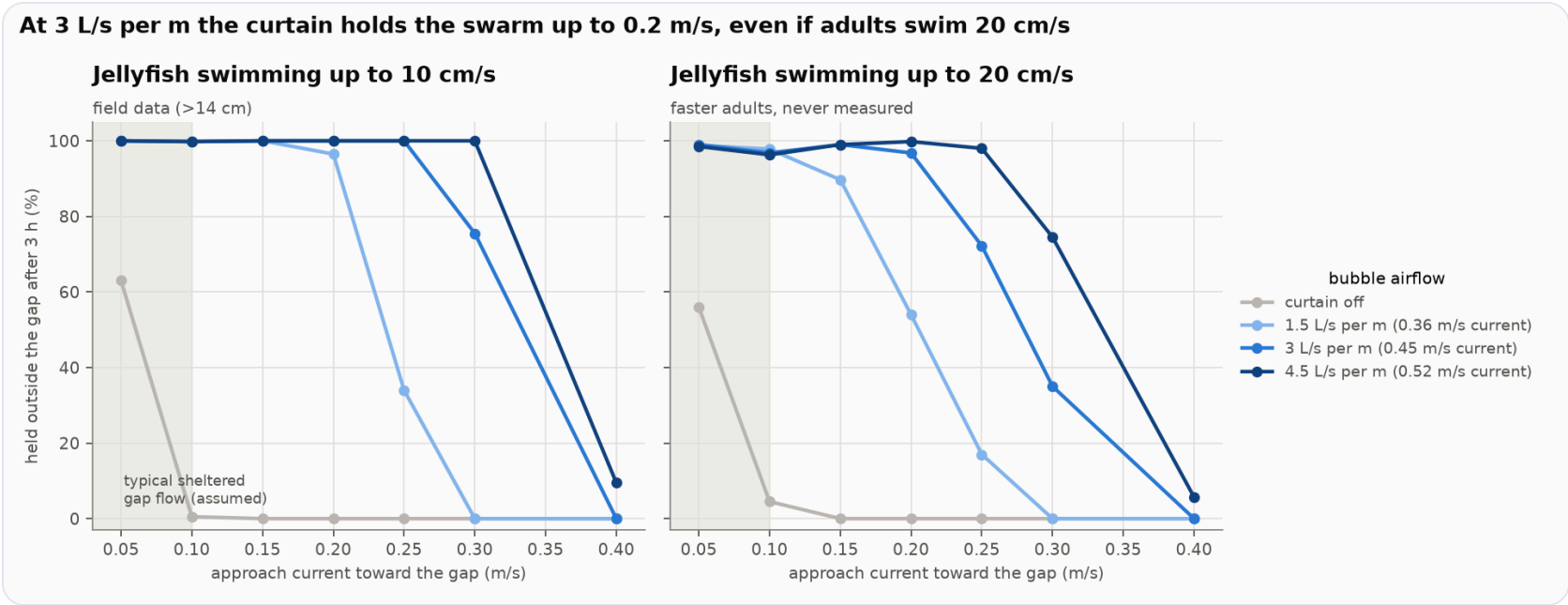
77% fewer hours

Versus switching on for every alert. A live forecast narrows the window further (10–77 h → ~10–20 h in tests).

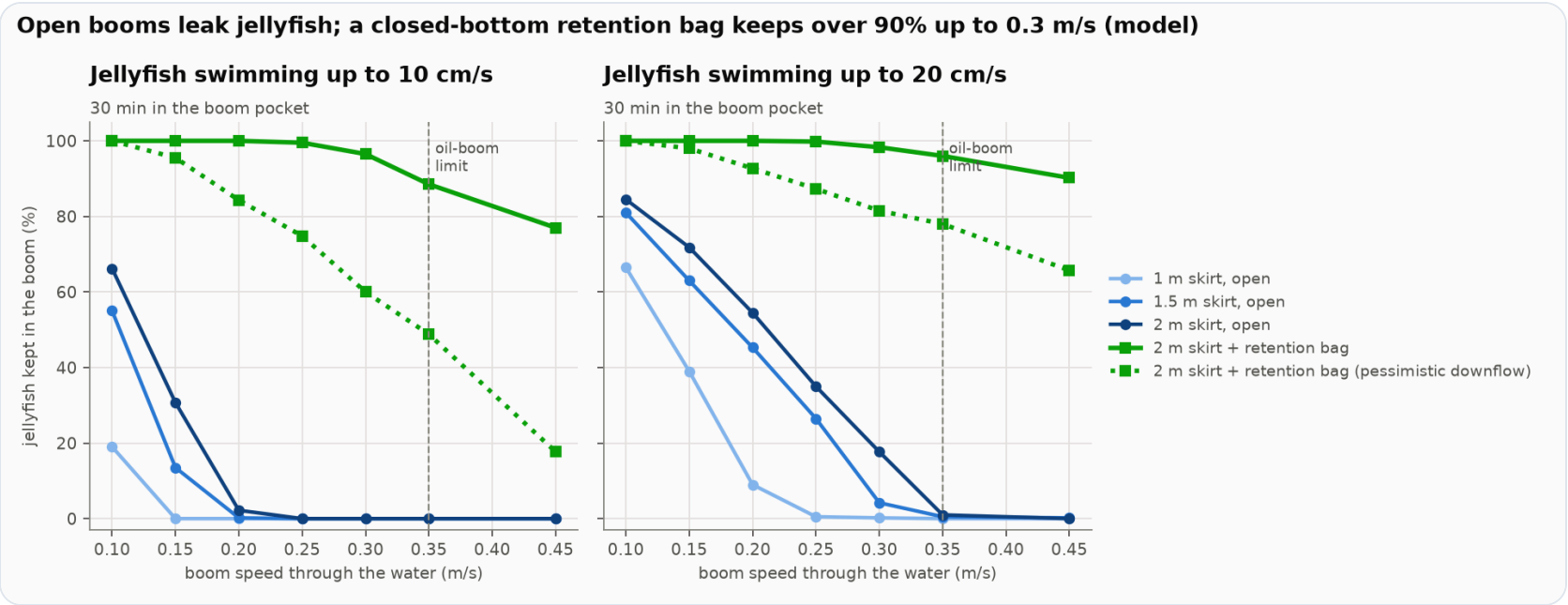
46% of the energy

Adaptive airflow follows real currents and wind. ~8% of hours need more air than the design maximum: flagged for the crew.

Our finding: oil-spill booms leak jellyfish, and a bag fixes it



400 individually simulated jellyfish per point; surface current from Bulson's pneumatic-barrier relation.



Jellyfish depth from field data; apex downflow 25% (pessimistic 50%) of tow speed.

Curtain holds the swarm

At 3 L/s of air per metre it holds 100% up to 0.2 m/s approach current, 95% even if adults swim 20 cm/s; real currents average 0.08 m/s. It fails above ~0.3 m/s, matching field trials.

Why the bag matters

Oil floats back up; neutrally buoyant jellyfish are dragged under the skirt far below the 0.35 m/s oil-boom limit (0–54% kept at 0.2 m/s). A closed-bottom fabric bag keeps 88–100% in the model.

Gentle on marine life, built from parts that exist today

Marine-life protection

RISK	HOW WE REDUCE IT
Jellyfish killed	Kept in the water and released alive; no cutting, pumping or mesh
Turtles, dolphins in the boom	Escape gap, slow tow, camera watch, stop on sighting
Seabed, sediment	Pipe on legs 0.5–1 m above the bed, routed away from coral and seagrass
Bubble noise, fish	Runs only during events; measured in the pilot
Released swarm returns	Release point simulated for 72 h; returners meet the curtain again (~1.3× load)
Carbon	~11 MWh per event: ~4.4 t CO ₂ on a gas grid, ~0.1 t on the plant’s own power

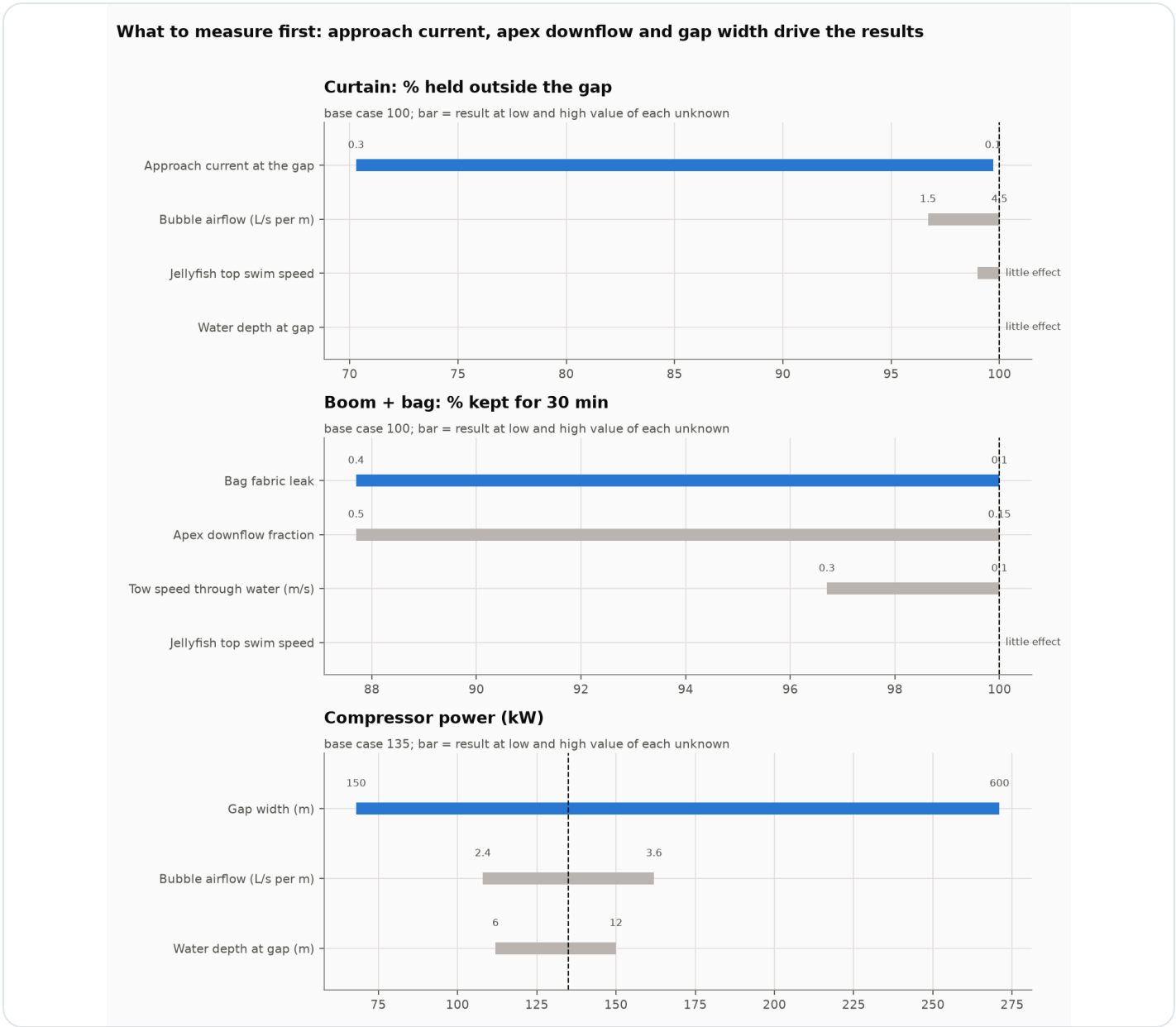
Feasibility

ITEM	ESTIMATE (300 M GAP)
Compressed air	0.9 m³/s, ~135 kW, oil-free (ISO 8573-1 Class 0)
Capital (indicative)	USD 0.4–1.3 M: curtain, compressor + standby, boom, bags, boats, sensors
Running cost	~USD 1,100 of energy per event
Value	One avoided 12 h unit outage (~USD 0.5 M, EPRI figure) pays back a third to all of it
Capacity	One boom pair: ~2,200–21,600 jellyfish/h (4–43 t/h) vs 0.9 t/h for a cutting robot
Gulf hardening	Membrane diffusers, foul-release coating, modular sections, lifted out off-season

What we assume, what we don't know yet, and what we measure first

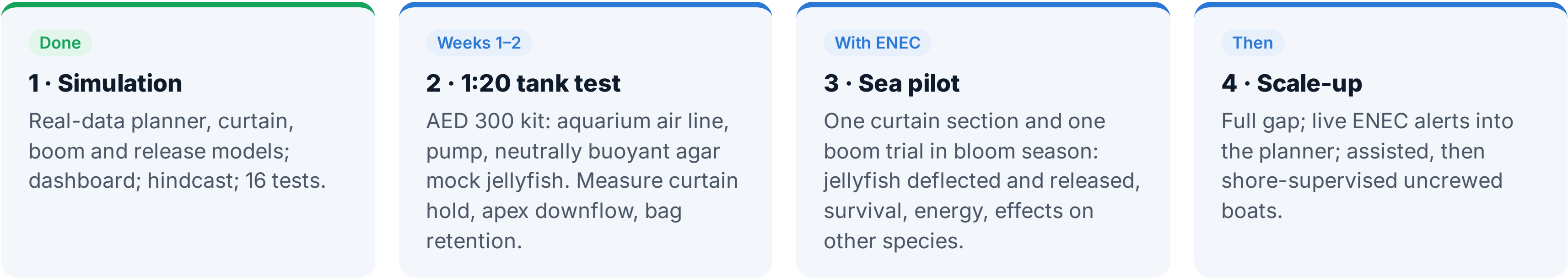
ASSUMPTION	BASIS	IF WRONG
Intake behind a breakwater with one gap, ~300 m, ~9.5 m deep	Brief; public channel depth	Curtain size scales; measure on satellite images
Alert gives position, hours of notice	Brief: "time to prepare"	Camera trigger switches the curtain on
Swarm in top 1 m; adults swim ≤10–20 cm/s	Field and lab studies (Australia)	Deeper skirt; tested at 20 cm/s
Swarm density 0.1–1 per m ³	No published figure	Capacity scales linearly
Apex downflow 25–50% of tow speed; bag passes 20%	Model hypothesis	Measured in the tank test

Limitations: no management method has been tested on Blue Blubbers; bubble curtains weaken in strong currents (~8% of hours need extra air); ~23% of released jellyfish may drift back within 3 days; uncrewed boats are a later phase.



Sensitivity: approach current, bag leakage / apex downflow and gap width drive the results, so we measure them first.

From simulation to sea, one measured step at a time

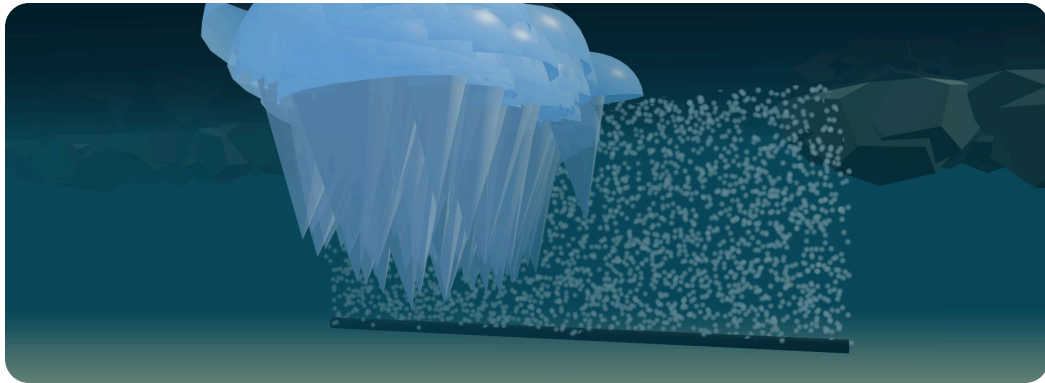
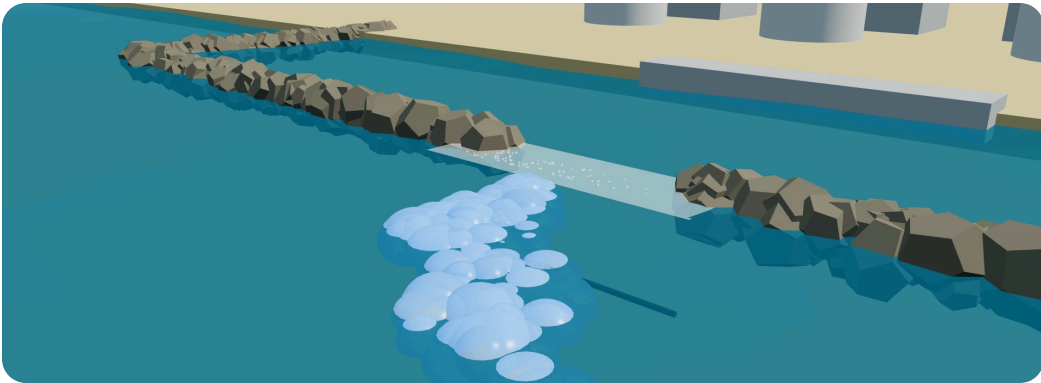


Pass marks

- Tank: $\geq 95\%$ held at the scaled 0.2 m/s current
- Tank: bag keeps $\geq 90\%$ at the scaled 0.2 m/s tow
- Pilot: no harm to non-target species; fewer screen-clearing hours

Open work and AI use

Code, data and tests: github.com/Shaaha-7/ENEC. AI assistants helped search and summarise literature, check claims, draft text and write simulation code; the team reviewed every decision, assumption and figure, and can explain each in our own words.



Stills from our 3D explainer video: the curtain holding a swarm outside the gap, and the bubble wall seen underwater.